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RESEARCH PROPOSAL

Full Legal Name: Lynn Thant Maung
Institution: IIP International School
Category: Junior
Class/Grade/Year: Grade 9 (Freshman)
Country: Myanmar
Submission Date: September 20, 2026
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Research Topic: Littering in Yangon's Flea Markets: A Mixed-Methods, Decision Theoretic Study of Vendors' and Shoppers' Waste Disposal Choices

Title: Littering in Yangon's Flea Markets: A Mixed-Methods, Decision-Theoretic Study of Vendors' and Shoppers' Waste Disposal Choices

Research Problem

Myanmar is in the medium human development category with an HDI of 0.609 as of the 2023 dataset [1]. Like many other developing countries, it struggles with waste management. Kyaw [2] points out the lack of awareness of proper trash disposal and several easily-fixable reasons why people litter on roadsides as shown by their surveys. Many Burmese citizens rely on roadside shops and flea markets to buy groceries and supplies, given the country's developmental status. These places tend to accumulate waste from the actions of the shopkeepers and the shoppers. This raises questions of why littering is so prevalent in these vital socio-economic zones and what can be done to reduce it. This study addresses this question by observing citizens' waste disposal decisions in Yangon's flea markets and testing whether these decisions follow the predictions of a decision-theoretic model based on norms, observability, identifiability and crowding.

Existing Literature

The original work done by Cialdini *et al*. [3] established the focus theory of normative conduct, which distinguished 'descriptive' and 'injunctive' norms, and that a person's decision to litter depended on which norm was more 'salient' in the situation. Krecik [4] and Rangoni and Jager [5] expanded upon this by using differential equations and agent-based modelling respectively, with the former focusing on feedback dynamics and the latter focusing on littering and adaptive cleaning strategies. Finally, Bergquist *et al*. [6] replicated Cialdini *et al*.'s experiments to confirm that context was a significant factor. These sources will be vital in the decision-theoretic model as variables to be included. Moving onto the Myanmar specific sources, Nesheim *et al*. [7] looked at the roles of scrap dealers, waste pickers, etc., shedding light upon the informal waste economy. Mon [8] described waste and informal jobs in Myanmar regarding managing it. Myanmar Digital News [9] described from a more personal perspective, highlighting the aspects of waste management in the city. Finally, Han [10] and Kyaw [2] both provided meaningful statistics to show data on the awareness of the citizens of Yangon. These local findings about waste management provide meaningful perspectives and will help interpret the qualitative findings in the interviews to be done after the field observation study.

The main research gaps are the lack of specialization in Yangon's flea markets, and the lack of mathematical modelling of littering decisions. This study acknowledges this gap and in addition, adds observability and identifiability as factors that may affect littering in social crowds.

Research Question

The **main research question** is:

To what extent does the norm-focus theory, observability and identifiability of the litterer, and the magnitude of crowding affect littering by vendors and shoppers in Yangon's local flea markets?

The **secondary research questions** are:

1. How do descriptive and injunctive norms influence littering decisions in Yangon's flea markets?
2. How do observability and identifiability moderate these effects?
3. Does crowding have a non-monotonic effect on littering behaviour?

The **hypotheses** are:

1. Higher descriptive norms and lower injunctive norms will lead to more littering.

2. Vendors, due to low perceived social sanction and a lack of consequences, will litter more than shoppers.
3. Crowding will have a non-monotonic effect on littering.

Methodology

This study will use a mixed-methods research design. The quantitative aspect will consist of observing and collecting data on disposal opportunities and the qualitative aspect will consist of interviewing a sample of shoppers and vendors alike to review their opinions and awareness of proper trash disposal.

A person will litter when the benefits of littering outweigh the costs, hence it is to be modelled as a decision-theoretic choice. This theoretical equation predicts that a person will litter when:

$$\mathbf{U(Litter) = i + d + o + s + v > 0}$$

Littering is the binary outcome, coded as yes (1) or no (0). The predicting variables are the inconvenience of proper disposal (i), estimated as distance from the nearest bin, coded as less than 10 meters (-1) or 10 meters or further (1). Descriptive norm (d) measured as the count of visible litter pieces within $10m^2$ of the person, coded as less than 5 (-1) or 5 or more (1). Observability (o), is coded as high visibility, e.g., main paths (-1) or low visibility, e.g. behind stalls (1). Observable injunctive cue (s) is coded as signage present (-1) or not (1). Vendor status is coded as vendor (1) or shopper (-1). Crowding is measured as the count of people within $10m^2$ at the time of observation and will be tested separately in the regression formula with a quadratic term.

Some predicting variables are coded as -1 or +1 so that the utility is increased or decreased, which matches the theoretical ideas of each variable. Vendor status is used as a predictor as they are more identifiable than shoppers. However, they also face fewer consequences for littering, which increases it. According to hypothesis 2, the net effect is that the vendors do litter more. Crowding is not included in the utility equation as it is hypothesised to be non-monotonic, only to be included in the regression model. As the equation is only a theoretical guide, it is subject to change in the pilot period. Not all terms can be measured in the field observation, so it tests a reduced-form logistic regression using the predicting variables.

Regression formula:

$$\log\left(\frac{P(\text{Litter} = 1)}{1 - P(\text{Litter} = 1)}\right) = \beta_0 + \beta_1 i + \beta_2 d + \beta_3 o + \beta_4 s + \beta_5 v + \beta_6 \text{Crowding} + \beta_7 \text{Crowding}^2 + \beta_8 (d \times o) + \beta_9 (d \times v)$$

The coefficients will test whether observability and vendor status moderate the effect of the descriptive norms. Market, zone, and time fixed effects will be included, and the significance level will be set at $p < .05$.

During the pilot week, observers will independently code a subset of events to check inter-rater reliability in data recording. A total of four adult observers, two at each market, will position themselves in two flea markets (one large and crowded, the other small and sparse) to note down any disposal opportunities and record all variables relating to them. A disposal opportunity begins when a person has waste to discard. The observers will record the outcome (littering vs. bin) and all the variables relating to it. Data will be collected over a three-week period. After the data collection stage, there will be interviews with local customers and vendors (n=30) on their opinions on littering with these 3 questions (all using the Likert Scale):

1. “How often do you see litter in this flea market?”

2. “How much do your personal beliefs affect how you dispose of trash?”
3. “How much do other people’s actions affect how you dispose of trash?”

Before data collection, permission from local authorities and the ethics review board will be obtained. Anonymity and verbal consent are to be maintained, with participants being able to stop at any time. No personal details such as names, faces, photos or videos will be taken. Data will be stored securely and destroyed after the study.

The study predicts descriptive norms affect littering more than injunctive norms. Vendors are expected to litter more than shoppers despite being more identifiable, due to lower consequences. Crowding is expected to have a U-shaped effect, with littering lowest at moderate crowding.

The study is limited to two Yangon flea markets, so findings may not generalize to other markets or cities. Only correlation is established, not causation. There is difficulty in measuring crowding and litter density precisely in the field.

Phase 1: Training observers and pilot will take 1 week.
Phase 2: Data collection will take 3 weeks.
Phase 3: Data analysis and write-up will take 2 weeks.
Phase 4: Evaluation and revision will take 1 week.

Research Topic

Yangon’s flea markets are vital for the Burmese people, specifically those with low-income [7]. They provide cheap prices and serve as social hubs. With the addition of litter, the local economy will be harmed in sales with customer dissatisfaction regarding sanitation. This may lead to sanitation risks for vendors and shoppers. Current information campaigns are failing because they try to make injunctive norms more visible. But in flea markets, descriptive norms (what people see others doing) are much more salient [3]. This context matters in terms of norm focus theory [6]. By identifying the factors that motivate littering decisions, the study can inform solutions to target those factors directly, rather than assuming the lack of information is the issue.

Quality of Writing

The language of the proposal is clear and direct, appropriate for a multidisciplinary audience. Technical terms are defined when introduced. The structure follows standard research norms and adheres to IEEE citation and formatting guidelines to meet IARCO standards.

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